Problem Chosen	2024 HiMCM/MidMCM Summary Sheet	Team Control Number
A		16159

Olympic Makeover: Modeling the Future of the Games

Every year, more than 3 billion people around the world tune in to the Olympic games. They watch events ranging from breakdancing to table tennis to triathlons, and for weeks and weeks, the Games are the only topic people bring up. For athletes, participating in the Olympics is the achievement of a lifetime, and it is regarded as the highest level of competition in the world. At times, the Olympics loom so large that it seems to include every sport imaginable. But when we look closer, there are several notable sports missing from the Games, and deciding what sports to include each year is no easy task.

For many years, it was very rare to see new sports gain access to participation in the Olympics. However, in the 2020 Tokyo Olympics, skateboarding, surfing, sport climbing, and several more sports were added to the roster, and this trend has continued since. The IOC's shift towards modernizing the selection of events makes it more important than ever to create an organized way to determine which sports to include. Thus, our team of HiMCM Olympic Consultants is here to provide an objective model which facilitates sport inclusion processes for the IOC.

Our model has three main parts: quantifying factors, normalizing variables, and creating entropy weights. The first step was to quantify the the different factors for each sport in order to provide an objective input on how well they satisfy the IOC's values.

After quantifying the factors, we applied the process of min-max normalization. This expressed all of our variables as indices in the range [0,1]. With the values now in the same interval, we checked for correlation in order to ensure that no variables were prioritized unfairly in our model.

Then, we used the entropy weight model to create an objective weighting for the variables that limited the effects of dispersion and uncertainty in our data. We used these weights on our normalized sample data to calculate scores for each sport. From there, we used the score for each sport to create a ranking of which ones suited Olympic values best, and used this list to suggest recommendations for sports to be included at the 2032 Summer Olympics.

Finally, we performed sensitivity analysis on our model to evaluate how it holds up against change. We analyzed our results and established several strengths and weaknesses of our model. These findings allowed us to conclude that our model is robust and objective, so we present our rankings to the IOC with confidence.

Contents

1 Introduction

In 1900, the first and only occurrence of pigeon shooting as an Olympic sport was won by a Belgian man, marking his place in the throne room of Olympic medalists with 21 pigeon kills. The runner-ups followed closely, with the fourth-place winner killing 18 pigeons.

It is to no one's surprise that this sport never again participated in the Olympics; the IOC deemed it too cruel (the fowl may have a few foul words regarding this). A number of factors pointed to the removal of this sport (and subsequently the IOC deemed it to be a demonstration sport rather than an official Olympic sport at the time), and while it is easy to point to the litter of pigeon carcasses and suggest it was too difficult for the crowd to process, or even too much work for the janitors to clean up, it was also a result of the times: the rise of animal activism at the time guaranteed that this sport would not return to the Olympics.

In 1908, the French decided to reinstate their monarchy in a rather unexplored area: the Olympics. The first and only instance of Jeu de paume, also known as royal tennis in less monarchical languages, seemed like a brilliant idea for a new Olympic addition: the origins of tennis as its own sport, elevated to a golden podium. Every point that a player received was a promising sight; no birds falling out of the sky, at the very least. However, its inevitable declining popularity (partially due to its complex scoring system) led to it being removed and never returning to the Olympics.

Perhaps it was at this time, with the International Olympics only a few years old, where the IOC came to an epiphany: some restrictions on sports would be beneficial. They developed six major categories that a sport must fall under to even be considered for the Olympics: the popularity and accessibility of the sport, its gender equity, sustainability, relevance and innovation, and safety and fair play.

Most sports that were removed and never returned to the Olympics had been added in the early 1900s, shortly after the establishment of the International Olympics and before any strict IOC guidelines. The IOC generally has high standards for the sports it accepts as Olympic sports, or even demonstration sports. But, in an ever-diversifying world, this can lead to problems: it is easier than ever for sports to be spread worldwide and fulfill the conditions set by the IOC.

Our model, designed to address these issues, takes into account all the aforementioned qualifications expected by the IOC. Normalizing the data to get rid of units, finding correlation between indices, and finally using an entropy weight model gave us scores for every sport that can be used to determine whether a sport should be considered for the Olympics.

2 Assumptions and Justifications

Assumption 1: We assume that characteristics of sports in practice for athletes are representative of characteristics of sports at the Olympics.
Justification: Since Olympic sports are a reflection of how sports are practiced in general, this is a fair assumption to make. This assumption is also necessary because there is no data on certain aspects

of sports at the Olympics, so we must supplement with data from general studies on sports.

Assumption 2: We split up the IOC's Fair Play and Safety criteria up into two separate factors of Fair Play and Safety.
Justification: We used different data to quantify the two parts of this IOC criteria, which made it difficult to treat the criteria as a single index. While analyzing the two data sets, we also discovered that the numbers we used to quantify Safety and Fair Play shared very little correlation (see Figure 4).

Assumption 3: We assume that all national/territorial member federations of the Union Cycliste Internationale participate in all of its disciplines, including E-Cycling.
Justification: This assumption was necessary because we were unable to find separate data on national federations of E-Cycling.

Assumption 4: We assume that the annual carbon footprint of a tug of war athlete is 514 kg CO_2-e.
Justification: According to Figure 1, carbon emissions for team sports derive mostly from travel for regular practice, competitions, league, and vacation. Since tug of war has a competition league, all of these factors will contribute to its carbon footprint as well. We were unable to find any additional information about emissions of a tug of war athlete in these categories, so we assumed that they would have the same carbon footprint as the average team/racket sport athlete. Figure 1 tells us that this value is 514 kg CO_2-e, so we used this for our data calculations.

Assumption 5: We assumed that data taken from a small number of countries is representative of global data.
Justification: Since the Olympics is a global event, we did our best to acquire data that included all countries when evaluating the impacts of our sports. However, there were multiple instances where we were unable to do this. Instead, we were only able to find relevant data from a single country (usually the US) or a few countries, so we worked under the assumption that distributions of a subset of countries would reflect the global distribution. For example, we considered Google trends while evaluating the Relevance/Innovation index for various sports, but many countries do not use Google. Although different countries have different norms and values that impact the factors we consider, this difference should not have a huge impact on our data.

Assumption 6: We assume that factors aside from Relevance/Innovation do not experience statistically significant changes over a few years.
Justification: There is no reason for a factor to change significantly for any sport over a couple of years, unless it is a factor like Relevance/Innovation that is specifically about recent changes. It is not unlikely that changes will accumulate over time, however, so we sourced data from recent years as much as possible.

3 Factors of our Model

3.1 Defining our Variables

Factor	Input	Normalized Index	Entropy	Weight
Popularity and Accessibility	p_i	P_i	E_1	W_1
Gender Equity	g_i	G_i	E_2	W_2
Sustainability	s_i	S_i	E_3	W_3
Inclusivity	n_i	N_i	E_4	W_4
Relevance and Innovation	r_i	R_i	E_5	W_5
Safety	a_i	A_i	E_6	W_6
Fair Play	f_i	F_i	E_7	W_7

The above table includes all of the variables for sport number i. For each of the seven factors, the lowercase input denotes whatever data value our model takes in for the ith sport. This gets converted into our uppercase index after being normalized.

Our normalized indices range from 0 to 1 and represent how close the sport is to meeting the IOC's standards. In other words, an index of 0 means a sport performs terribly in a certain category while a 1 means the sport does well.

Finally, we calculate entropy for each factor's indices, which we then use to determine how much weight each factor gets in consideration of our final score. These variables are the same for all sports, so the index subscript denotes which factor it applies to.

3.2 Popularity and Accessibility

The first factor the IOC considers for Olympic selection is how popular a potential sport is. Sports that get included in the Olympics should increase global interest in the Games and attract higher viewership. Thus, The sport should be widely played and easily accessible to people across the globe. This ensures that the sport is available to many people and not just played by a small group or region.

To find out whether a sport is popular and accessible or not, we assembled data on how much of the world participates in the sport. Thus, p_i is equal to the percentage of the world population that plays sport number i.

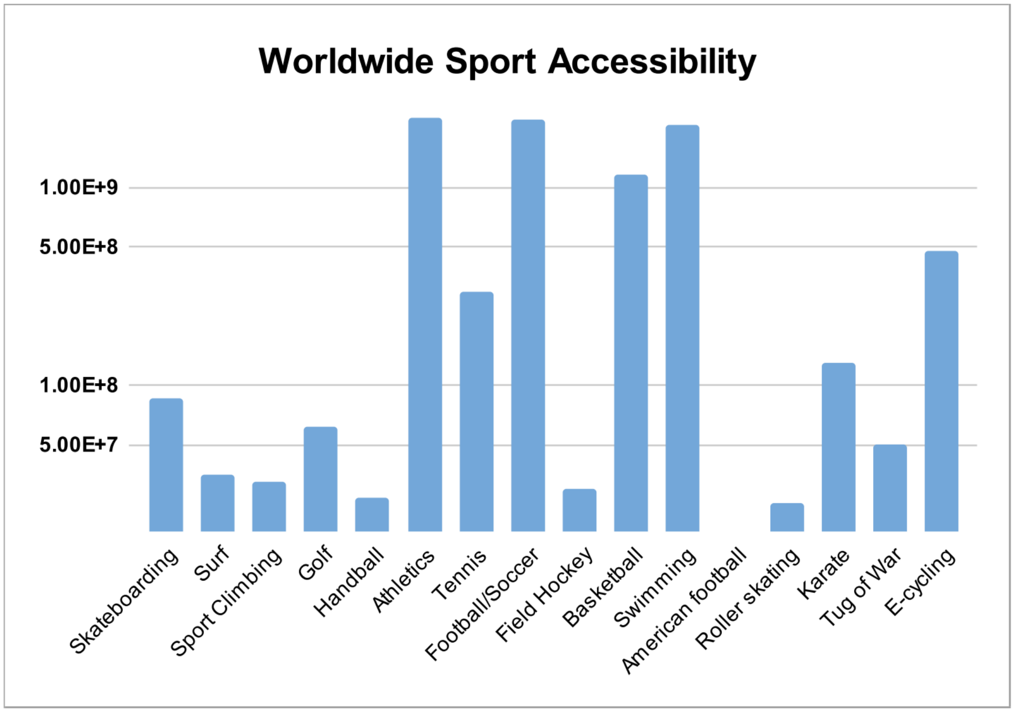

Figure 1: People in the world who play each sport

3.3 Gender Equity

The IOC also wants to ensure that the Olympics are inclusive of both genders. Thus, the sports added should be somewhat equally played among men and women.

To quantify gender equity, we found data regarding the percentage of female and male athletes in each sport. Let μ_i be the percentage of sport i players who identify as male and let γ_i be the percentage who identify as female. We want to minimize $|\mu_i - \gamma_i|$, which is the difference between the percentages of males and females. However, since this calculation rewards large disparities, we let $g_i = -|\mu_i - \gamma_i|$. This way, a large disparity between male and female participation will result in a lower index rather than a higher one.

3.4 Sustainability

The environmental impacts of hosting the Olympic games are at the forefront of the IOC's concerns. In order for a sport to be added to the Olympics, it is important to first evaluate what further environmental impacts for the host city it will cause.

Table 5
Annual total and partial carbon footprints by sport (mean values in kg CO_2-e).

Sport	CF total	CF regular	CF competition	CF league	CF day trip	CF vacation
Climbing/bouldering	1156.0	634.0	7.0	–	119.3	399.6
Diving	2,840.7	114.9	–	–	134.5	2591.3
Figure/roller skating	1238.4	686.6	435.6	–	15.9	100.3
Fitness (gym)	227.9	159.1	9.1	–	–	59.7
Golf	2195.0	565.6	161.9	–	93.4	1374.2
Headis	267.1	99.0	168.1	–	–	–
Hiking/walking	901.1	396.3	–	–	84.5	420.3
Skateboarding	717.9	373.8	33.2	–	105.3	205.6
Surf sports	2074.3	176.2	45.5	–	365.7	1486.8
Swimming	622.7	313.1	165.9	–	–	143.7
Track and field	361.5	134.8	155.7	–	–	71.0
Triathlon	774.8	183.1	282.1	–	–	309.7
Individual sports	1006.5	363.4	103.2	–	123.1	503.7
Nature sports	1455.2	477.5	12.2	–	139.6	829.5
American football	842.0	666.6	11.0	140.8	–	23.7
Basketball	681.3	441.7	36.6	198.1	–	4.9
Field hockey	874.2	529.5	71.6	235.0	–	38.2
Football (soccer)	337.4	230.7	19.5	74.3	–	13.0
Handball	405.4	218.6	19.1	155.8	–	11.9
Table tennis	392.8	232.3	46.6	103.4	–	10.5
Tennis	243.1	156.3	31.7	33.8	–	21.3
Volleyball (indoor)	404.8	248.3	51.6	95.4	–	9.5
Team/racket sports	514.0	342.4	33.6	121.4	–	16.6
Full sample	844.0	356.4	77.9	121.4	123.1	337.8

Figure 2: Carbon Footprint of Various Sports

We found that the most quantitative and objective way to judge a sport's sustainability is through its carbon footprint. Thus, s_i is equal to the negative of the carbon footprint (kg CO_2-e) of a single player of the ith sport per year.

3.5 Inclusivity

In order for a sport to be included in the Olympics, it must involve participation from countries across the globe. To calculate our inclusivity scores, we used the number of national federations of a sport.

According to the Olympics website, any included sport must be practiced in at least 75 countries across 4 continents. Thus, if a sport does not satisfy this requirement (has less than 75 national federations or covers less than 4 continents), our model returns that it cannot be an Olympic sport and it is removed from any further data pool calculations. Otherwise, n_i is the number of national federations of a sport.

3.6 Relevance and Innovation

According to the IOC's Olympic values, this factor should take into account 3 qualities: a sport's appeal to younger audiences, its reflection of modern trends, and its incorporated innovations (such as physical virtual sports). We claim that there is a correlation between each value and the trend of the number of Google searches the sport receives.

First, younger audiences are more likely to search up information about the sport because they use

Google more compared to older fans. Additionally, Google search trends clearly reflect how modern trends relate to a sport, and new innovations in a sport will lead to a spike in searches due to press releases and other media. Thus, we used Google search trends of various sport names to quantify their Relevance and Innovation factors. Google also allowed us to isolate searches for the sport instead of all searches of the word, so our data is limited to the scope we prefer.

Taking a look at the data, we notice that there are spikes for every sport once a year. This can be attributed to the fact that most sports are seasonal, making them most popular during one point in the year. After locating the yearly maximum, we perform linear regression on the points to establish the slope of the best-fit line. This gives us an approximate figure for the rate at which a sport grows, so this is the value we use for r_i.

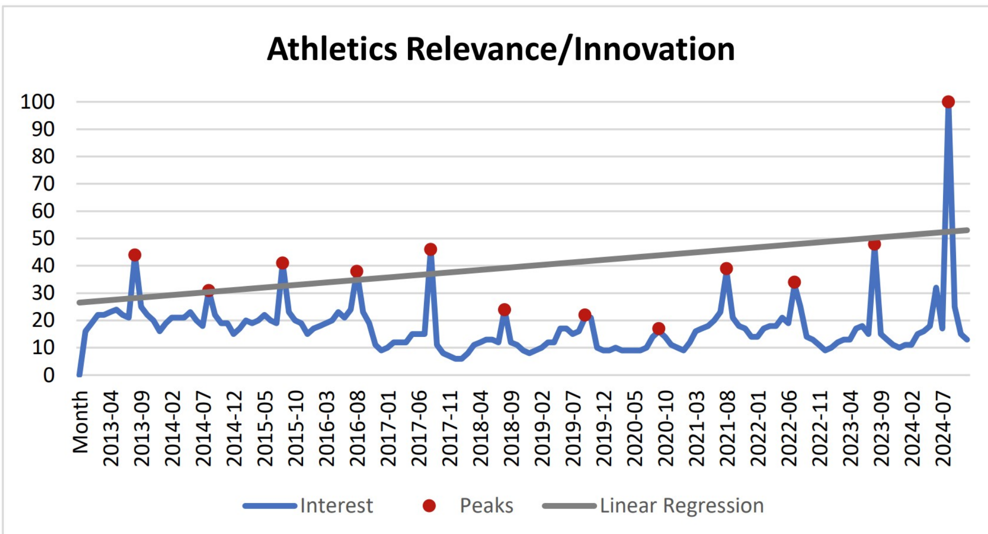

Figure 3: Relevance and Innovation Graph for Athletics

3.7 Safety

This Olympic value emphasizes the importance of protecting the safety of athletes and preventing injuries. To quantify this value, we researched the number of injuries an athlete of each sport receives. However, the larger this value is, the lower of a score the sport should receive, so we flip the sign. Thus, a_i equals -1 times the average number of injuries acquired per year by $100,000$ athletes playing sport i.

3.8 Fair Play

Finally, it is important for all sports included in the Olympics to promote fair play among competitors. Practically, this means that sports should have a low number of annual doping cases. However, we needed to account for the fact that sports with more players will naturally have higher amounts of doping cases. Additionally, we need to negate the number of doping cases so that our model will give a higher score to sports with low doping rates.

To deal with this, we let f_i be -1 times the number of reported doping cases of the sport in 2020 divided by p_i, which is the percentage of the world that plays sport i. This accounts for the correlation between the number of doping cases and popularity, thus giving us a f_i that is affected solely by doping counts.

4 Min-Max Normalization

Now that we have all of our variables, we need need to apply a normalization process to produce indices that we can work with. This is necessary because all of our variables currently have different units of measure and orders of magnitude. In order to standardize all of them and highlight the relationship between data values, we will apply a min-max normalizaton method to all of our variables.

For example, consider all of our p_i. We want to normalize all of our values into P_i indices, which should be distributed from 0 to 1. To do this, we let

$$P_i = \frac{p_i - min(p_i)}{max(p_i) - min(p_i)}$$

We apply the above min-max normalization method to all $p_i, g_i, s_i, n_i, r_i, a_i$, and f_i to achieve a normalized index for each sport in each category. This preserves the distribution of our data while scaling it to fit the interval $[0, 1]$, which is important because it lets us work with values of varying units.

5 Correlation Analysis

Next, we performed correlation analysis on our normalized indices to see if any factors are correlated with each other. The heatmap below displays the correlation coefficient between every pair of factors and the highest ones are colored red.

	Accessibility	Gender	Sustainability	Inclusivity	Relevance/ Innovation	Fair Play	Safety
Accessibility	1.00	-0.02	0.45	0.62	-0.13	0.31	-0.25
Gender	-0.02	1.00	0.04	0.39	0.04	0.06	0.65
Sustainability	0.45	0.04	1.00	0.55	-0.11	-0.13	-0.33
Inclusivity	0.62	0.39	0.55	1.00	-0.16	0.12	-0.11
Relevance/ Innovation	-0.13	0.04	-0.11	-0.16	1.00	0.06	0.05
Fair Play	0.31	0.06	-0.13	0.12	0.06	1.00	0.17
Safety	-0.25	0.65	-0.33	-0.11	0.05	0.17	1.00

Figure 4: Correlation Matrix

We can see that the highest correlation between two different factors is 0.62, which is not statistically significant enough to consider accessibility and inclusivity correlated. Thus, since none of our factors are correlated, there is no uneven weighting present in our model so we do not need to remove any factors.

6 Entropy Weight Model

With these indices in mind, we aim to calculate a total score representing a sport's overall suitability for being included in the Olympics. To achieve this, we must weigh the various factors involved in this calculating this score. After considering various models, we decided to employ the Entropy Weight Model. This model uses the dispersion of values to determine their reliability in decision making.

6.1 Priming the Data Matrix

We begin with a matrix of index values, with each row containing the data of a sport and each column containing the data of one of the 7 factors, as displayed using our sample data in figure 5.

	Accessibility	Gender	Sustainability	Inclusivity	Relevance/ Innovation	Fair Play	Safety
Skateboarding	0.030053	0.311828	0.716691	0.405405	0.700683	1.000000	0.344903
Surfing	0.007548	0.698925	0.058564	0.277027	0.594791	0.845714	0.977399
Sport Climbing	0.006418	0.475269	0.504124	0.168919	1.000000	0.833792	0.999281
Golf	0.019341	0.655914	0.000000	0.500000	0.458582	0.911764	0.824225
Handball	0.003948	0.806452	0.868316	1.000000	0.661827	0.000000	0.581438
Athletics	1.000000	0.903226	0.889617	0.939189	0.599488	0.828836	0.956559
Tennis	0.125221	0.956989	0.947065	0.972973	0.654996	0.927134	0.809637
Football/Soccer	0.979177	0.182796	0.901310	0.918919	0.000000	0.903998	0.373037
Field Hockey	0.005298	1.000000	0.640854	0.432432	0.076430	0.640000	0.906543
Basketball	0.516585	0.320430	0.734449	0.864865	0.606746	0.939791	0.000000
Swimming	0.915415	0.606452	0.762882	0.972973	0.765585	0.976316	0.516834
American Football	0.000000	0.000000	0.656477	0.000000	0.444065	0.111330	0.213611
Roller Skating	0.003221	0.268817	0.464338	0.405405	0.514518	0.787275	0.816931
Karate	0.050308	0.612903	0.811289	0.777027	0.183604	0.916923	0.908419
Tug of War	0.014370	0.500939	0.627802	0.006757	0.710931	0.892336	0.688869
E-Cycling	0.207843	0.655914	1.000000	0.905405	0.547822	1.000000	1.000000

Figure 5: Preliminary Index Matrix

Let the terms of this matrix be denoted as a_{ij} where

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,7} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,7} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,7} \end{bmatrix}$$

We first normalize the terms by dividing each value by the sum of its column. This transforms terms in the $m \times 7$ matrix A into terms in matrix B, $b_{i,j}$ for which

$$b_{ij} = \frac{a_{ij}}{\sum_{i=1}^{m} a_{ij}}.$$

For our example matrix, the new normalized matrix looks as follows:

	Accessibility	Gender	Sustainability	Inclusivity	Relevance/ Innovation	Fair Play	Safety
Skateboarding	0.007736	0.034814	0.067716	0.042463	0.082239	0.079903	0.031591
Surfing	0.001943	0.078032	0.005533	0.029016	0.069811	0.067575	0.089524
Sport Climbing	0.001652	0.053062	0.047632	0.017693	0.117370	0.066622	0.091529
Golf	0.004979	0.073230	0.000000	0.052371	0.053824	0.072852	0.075494
Handball	0.001016	0.090037	0.082042	0.104742	0.077679	0.000000	0.053257
Athletics	0.257417	0.100842	0.084055	0.098372	0.070362	0.066226	0.087616
Tennis	0.032234	0.106844	0.089483	0.101911	0.076877	0.074081	0.074158
Football/Soccer	0.252057	0.020408	0.085160	0.096249	0.000000	0.072232	0.034168
Field Hockey	0.001364	0.111646	0.060551	0.045294	0.008971	0.051138	0.083034
Basketball	0.132978	0.035775	0.069394	0.090587	0.071214	0.075092	0.000000
Swimming	0.235644	0.067708	0.072080	0.101911	0.089857	0.078010	0.047339
American Football	0.000000	0.000000	0.062027	0.000000	0.052120	0.008896	0.019566
Roller Skating	0.000829	0.030012	0.043873	0.042463	0.060389	0.062905	0.074826
Karate	0.012950	0.068428	0.076654	0.081387	0.021550	0.073265	0.083206
Tug of War	0.003699	0.055928	0.059317	0.000708	0.083442	0.071300	0.063097
E-Cycling	0.053502	0.073230	0.094484	0.094834	0.064298	0.079903	0.091595

Figure 6: Normalized Matrix

6.2 Computing Entropy

Put simply, statistical entropy is the measure of uncertainty or ”surprise” in the data. It tells us how ”random” or dispersed the data for each factor is, which we can use as a metric for the reliability. We calculate the entropy of each decision-making factor:

$$E_j = -\frac{1}{\ln m} \sum_{i=1}^{m} b_{ij} \ln b_{ij}, \, j = 1, 2, \ldots, 7$$

This gives us the values of entropy for each factor

	Accessibility	Gender	Sustainability	Inclusivity	Relevance/ Innovation	Fair Play	Safety
Entropy	4.510653934	6.713385119	6.762774262	6.521271067	6.718324867	6.82492237	6.769089012

Figure 7: Entropy Matrix

6.3 Calculating Weights

Now, using the entropies of each factor, we can determine how much each factor should be weighted in our final score. We calculate the weights W_j as:

$$W_j = \frac{1 - E_j}{\sum_{j=1}^{7} (1 - E_j)}, \, j = 1, 2, \ldots, 7.$$

This formula provides a higher weighting to factors with more concentrated data (low entropy) and a lower weighting to factors that have highly dispersed data (high entropy.)

	Accessibility	Gender	Sustainability	Inclusivity	Relevance/ Innovation	Fair Play	Safety
Weight	0.09282429638	0.1510661443	0.1523720299	0.1459865061	0.1511967549	0.154015272	0.1525389966

Figure 8: Weighting Matrix

The result is a set of weights $W_j, j = 1, 2, \ldots, m$ that satisfy $\sum_{j=1}^{7} W_j = 1$. We can now use these weights to calculate a final score for each sport. This score, Q_i is calculated by

$$Q_i = W_1 P_i + W_2 G_i + \cdots + W_7 F_i, i = 1, 2, \ldots, m$$

7 Results

7.1 Score Report

To test our model, we gathered data on 16 sports, some established, some recently added or removed, and some never done before. For our example data, this leaves us with a final weighting as shown below. Calculating all the indices and using the entropy model, we get the following result for scores:

Rank	Sport	Olympic Status	Score
1	Athletics	Established	0.8661394261
2	Tennis	Established	0.8078665377
3	E-Cycling	Never Done	0.7923114671
4	Swimming	Established	0.7798287827
5	Karate	Recently Removed in 2024	0.6418616665
6	Sport Climbing	Recently Added in 2020	0.6059099493
7	Handball	Established	0.5892457401
8	Football/Soccer	Established	0.5861220471
9	Basketball	Established	0.5710062451
10	Field Hockey	Established	0.5607444178
11	Skateboarding	Recently Added in 2020	0.5308511794
12	Surfing	Recently Added in 2020	0.524925103
13	Tug of War	Removed in 1920	0.5236578494
14	Golf	Reintroduced in 2016	0.5093631103
15	Roller Skating	Never Done	0.4945034255
16	American Football	Never Done	0.2169004862

Figure 9: Sample sport ranking

At a glance, this ranking makes sense. All of the established sports, which have been included in the Olympics for decades, ranked in the top 10 and received a final score above 0.5.

7.2 Existing Sports

To confirm the accuracy of our model, we check the scores of some established sports in the Olympics. Below, we include indices for the established SDEs athletics, tennis, and swimming, all three of which have been around since the 1800s.

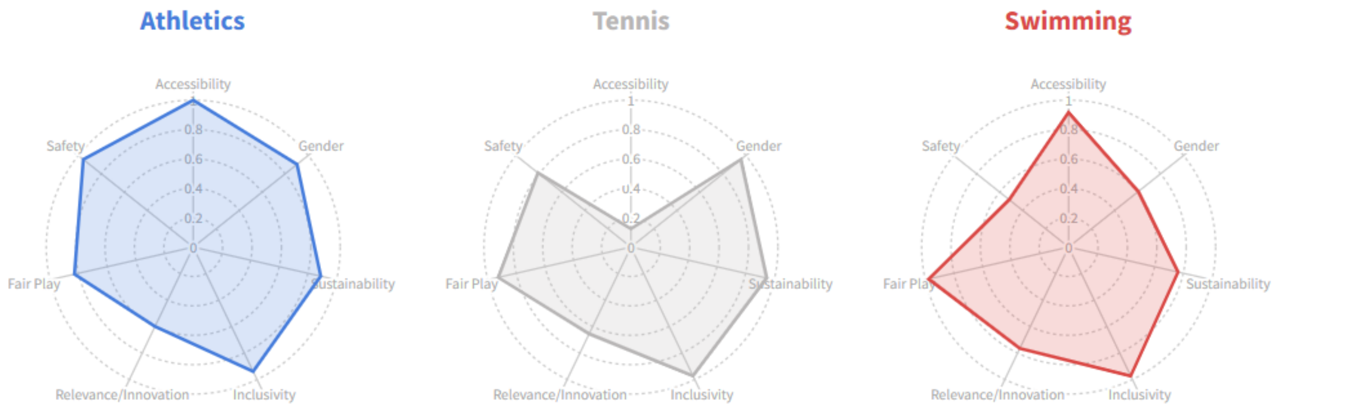

Figure 10: Indices of Established Sports

We notice that all three SDEs have very high indices. This makes sense because these are the sports the Olympic values are structured around, so they should score very well against their requirements. Since these sports do so well, it makes sense for the Olympics to keep them around year after year, and confirms the reliability of our model.

We will next investigate the results of our model when applied to the current status of some recently added or removed SDEs. Three examples are climbing, skateboarding, and surfing, all of which were first introduced in the 2020 Olympics.

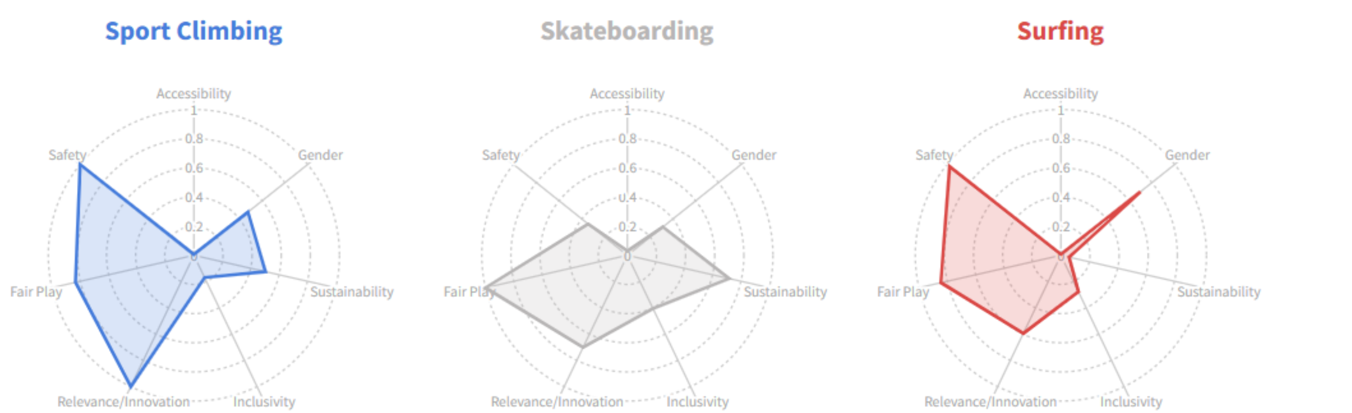

Figure 11: Indices of Newly Introduced Sports

While their values are not as high as established sports, they still have relatively large score values. Specifically, all 3 sports have very low accessibility values, which makes sense because newly established Olympic sports are likely not as widespread globally as ones that have been around for years. However, these sports score well in their other categories, which is likely why they were approved to join the Olympics.

7.3 Potential New Sports

Three SDEs that the IOC should consider adding to the 2032 Summer Olympics in order of highest priority are E-cycling, Karate, and Tug of War, respectively.

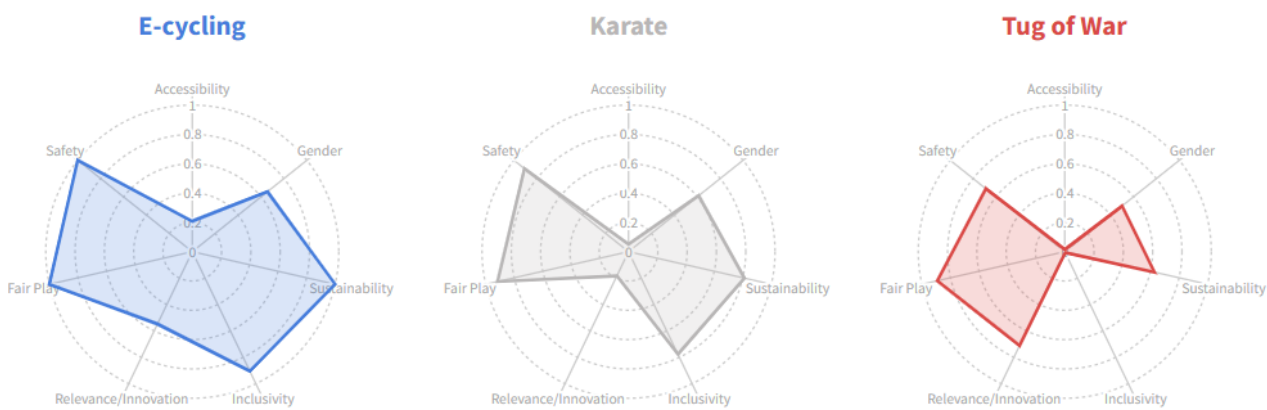

Figure 12: Proposed SDE indices

E-cycling had a relative high score of roughly 0.7923. It ranked higher in our model than many currently established SDEs, making it a clear choice for an addition to the 2032 Olympics. Karate ran for only one year in 2020, but our model suggests that it has the potential to be a very safe, sustainable, and cost-efficient sport. Tug of War was removed in 1920, but has the potential to be reintroduced with its relatively high relevance score and decently high score value.

Analyzing future trends for our sample sports, we predict roller skating could become a viable Olympic sport by 2036 or later. While accessibility is currently low, other factors (Figure 5) score relatively high, and the relevance/innovation factor of linear regression slope is 1.58 indicating growing interest. This growth is likely to improve accessibility and overall scores over time.

8 Analysis of our Model

8.1 Sensitivity Analysis

Now that we have our model and its results, we need to assess how robust our model is.

First, we decided to do testing on W_4 first because that was the largest weight, so we wanted to see how much impact a change would have on the overall rankings. We scaled weight W_4 by 120% and modified other W_j equally in order to preserve a weight sum of 1. Then, we ran our model again with our new weights and obtained a new ranking for our 16 sports. The comparison is shown below:

	Original Ranking		Sensitivity Test 1 (Inclusivity increase 20%)	
Rank	**Sport**	**Score**	**Sport**	**Score**
1	Athletics	0.8661394261	Athletics	0.8683652791
2	Tennis	0.8078665377	Tennis	0.8147609766
3	E-Cycling	0.7923114671	E-Cycling	0.7972791619
4	Swimming	0.7798287827	Swimming	0.7861273893
5	Karate	0.6418616665	Karate	0.6475975566
6	Sport Climbing	0.6059099493	Handball	0.6042240478
7	Handball	0.5892457401	Football/Soccer	0.5966972892
8	Football/Soccer	0.5861220471	Sport Climbing	0.5922584045
9	Basketball	0.5710062451	Basketball	0.5810850887
10	Field Hockey	0.5607444178	Field Hockey	0.5574620047
11	Skateboarding	0.5308511794	Skateboarding	0.5275824142
12	Surfing	0.524925103	Surfing	0.5175246616
13	Tug of War	0.5236578494	Golf	0.5099965657
14	Golf	0.5093631103	Tug of War	0.5071384724
15	Roller Skating	0.4945034255	Roller Skating	0.4924466409
16	American Football	0.2169004862	American Football	0.2099637756

Figure 13: Original vs. Post-W_4 Change

Next, we scaled W_1, our smallest weight, by 120% as well and scaled other weights to match. The new rankings are compared to the original ones below:

	Original Ranking		Sensitivity Test 2 (Accessibility increase 20%)	
Rank	Sport	Score	Sport	Score
1	Athletics	0.8661394261	Athletics	0.8688718175
2	Tennis	0.8078665377	Tennis	0.7938888391
3	E-Cycling	0.7923114671	Swimming	0.7825870566
4	Swimming	0.7798287827	E-Cycling	0.780361618
5	Karate	0.6418616665	Karate	0.6297687709
6	Sport Climbing	0.6059099493	Football/Soccer	0.5941513543
7	Handball	0.5892457401	Sport Climbing	0.5937101291
8	Football/Soccer	0.5861220471	Handball	0.5771960701
9	Basketball	0.5710062451	Basketball	0.5698714087
10	Field Hockey	0.5607444178	Field Hockey	0.549406015
11	Skateboarding	0.5308511794	Skateboarding	0.5206430114
12	Surfing	0.524925103	Surfing	0.5143829559
13	Tug of War	0.5236578494	Tug of War	0.5133190417
14	Golf	0.5093631103	Golf	0.4993552876
15	Roller Skating	0.4945034255	Roller Skating	0.4844847149
16	American Football	0.2169004862	American Football	0.2124898369

Figure 14: Original vs. Post-W_1 Change

Even though our scores were changed slightly, the overall ranking remained largely the same. A ranking that is very sensitive to weighting changes would be volatile and unreliable for decision-making. Since our results are consistent even under different weightings, our model withstands change well and is shown to be robust, so it can be applied for use in Olympics decision-making.

8.2 Strengths of our Model

1. Our model is objective. We avoided subjective processes wherever possible, quantifying all of our factors using data that we sourced rather than basing it off of any personal judgments. Our entropy model then allows us to have objective weights for each factor, which keeps our model unbiased.

2. Our model is not overly swayed by outliers. Through the Entropy Weight Model, we can determine the reliability and weighting of the factors based on the dispersion of the data. This allows our model to mitigate the effects of inconsistencies in input data, which makes for a more reliable result.

3. Our model makes it easy for sports to understand what they are missing from fitting the Olympic values. Since our model involves creating normalized indices for each factor, creating a web graph (see Figures 10, 11, 12) is very easy. These types of charts help sports federations quickly see what areas they do well in and what areas need improvement, which can be helpful if they hope to one day enter

the Olympics.

8.3 Weaknesses of our Model

1. A weakness of max-min normalization is that outliers can heavily impact the indices of other sports. if an extreme outlier is added to a dataset, it can make distinctions in the non-outliers less visible in the indices. This may seem like a critical flaw, but the effects of this are mitigated by the entropy weight model.

2. Another weakness of our model is the fact that it can only determine the suitability of a sport relative to other sports, and not independently. The only way for us to determine whether or not a sport is ”good” is by comparing it with other sports. So if a limited dataset is inputted, it may be difficult to draw conclusions from the ranking, since we have no definitive threshold for what is considered a ”good” or ”bad” score.

3. A concern we have about our model is that the results are not entirely consistent with the previous decisions of the IOC. Specifically, karate was included once in the 2020 Olympics and then removed from 2024 and 2028. We were surprised to see that karate was ranked so highly in our model's results, with a score higher than football/soccer and basketball. We are unsure why karate was removed, but this discrepancy could imply that the IOC has a different way of evaluating sports than our model does.

4. One more flaw is that this model cannot make long-term predictions. Since it's heavily dependent on current data, it is unable to project which sports will be suitable far into the future, or how the factors affecting sports will change over time.

9 IOC Letter

Dear International Olympic Committee,

We want to start off by expressing admiration for your work towards expanding and modernizing the events included in the Olympic Games. The new sports added in recent events, such as skateboarding and surfing, include some of our personal favorites to watch. In this process, we recognize how important adding and removing sports from the Games is, since it allows the Committee to determine whether or not a sport fits with the current program. However, we think it would be more efficient to apply an objective mathematical model with a more precise rating on how much a sport aligns with the values of your committee. For this reason, our team has created such a model, and we would like to present you with our findings.

It would be best to first introduce our model and explain its usage. Our model assigns a score to each sport using data about accessibility, gender distribution, sustainability, inclusivity, relevance and innovation, fair play, and safety. We designed it to determine how well a sport would fit into the IOC's guidelines and initiatives. Using these scores, we can then rank various sports, determining which ones are suitable to remove and which to keep. Additionally, we can apply our model to non-Olympic sports to decide if they are suitable to add to the roster.

We write to you with a proposal for the addition of 3 Olympic sports to the current roster: E-Cycling, Karate, and Tug of War, with priority in that order. We determined these priorities by applying our model using data gathered on a slate of 16 sports, which allowed us to compare potential new sports to existing ones. We explain each of our choices below, and hope you reciprocate with sufficient consideration.

E-Cycling, by our model, would be the best new addition as an Olympic Sport. With the abundant rise of new technology and the creation of an E-Sports division of the Olympics, it seems appropriate to start considering how physical-virtual sports can be integrated into the Games. Compared to normal biking, it is incredibly safe–it is not possible to fall off the bicycle (perhaps it is, but that would certainly be impressive), riders cannot ride into each other–and many other risks associated with biking are mitigated. It historically has lower doping rates than classic biking, and its sustainability is similar to that of biking: while more heavy equipment is required, contestants do not have to travel anywhere to practice, thus minimizing carbon emissions in this sense. Overall, E-Cycling is an excellent choice for inclusion in the upcoming Olympics Game, and it scores better in our model than several Olympic events that have been around since the start

Karate and Tug of War were both once Olympic Sports, but were eventually removed. Both sports are relatively safe and historically experience few doping cases. We found that these sports align well with many of your objectives and think they would be suitable additions as well.

Overall, we hope that these suggestions provide you with meaningful insights into our analysis. We look forward to seeing how these suggestions will support your efforts to modernize the Olympics through the inclusion of new and inclusive sports.

With kind regards,
HiMCM Olympic Consultant Team

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11 Appendix

Here is the code we used to generate indexes for our model:

```
import pandas as pd
import matplotlib as mpl
import numpy as np
import math
import statistics as stats

normalizedValues = [["Skateboarding (recently added)", "Surf (recently added)", "
                                        Sport Climbing (recently added)", "Golf
                                        (recently added)", "Handball (
                                        established)", "Athletics (established)"
                                        , "Tennis (established)", "Football/
                                        Soccer (established)", "Field Hockey (
                                        established)", "Basketball (established)
                                        ", "Swimming (established)", "American
                                        football (never done)", "Roller skating
                                        (never done)", "Karate (Recently removed
                                        )", "Tug of War (while back)", "E-
                                        cycling (new)"]]
```

```
def dataList(file):
    with open(file, 'r', encoding="utf8") as data:
        dataList = []
        for line in data.read().split('\n'):
            row = []
            for term in line.split(','):
                row.append(term)
            dataList.append(row)
    return dataList

def meanCol(lst, column):
  mySum = 0
  for row in lst:
    mySum += float(row[column])
  return mySum/len(lst)

def stdDevCol(lst, column):
  row = []
  for n in lst:
    row.append(float(n[column]))
  return stats.stdev(row)

def maxCol(lst, column):
  row = []
  for term in lst:
    row.append(float(term[column]))
  return max(row)

def minCol(lst, column):
  row = []
  for term in lst:
    row.append(float(term[column]))
  return min(row)

# POPULARITY & ACCESSIBILITY
accData = dataList("../PACK-HiMCM/PACK HIMCM DATA - Accessibility.csv")
accScores = [(float(n[1])-minCol(accData[1:], 1))/(maxCol(accData[1:], 1)-minCol(
                                        accData[1:], 1)) for n in accData[1:]]
normalizedValues.append(accScores)

# GENDER
genderData = dataList("../PACK-HiMCM/PACK HIMCM DATA - Gender.csv")
genderGaps = [-abs(float(n[1])-float(n[2])) for n in genderData[1:]]
genderScores = [(n-min(genderGaps))/(max(genderGaps)-min(genderGaps)) for n in
                                        genderGaps]
normalizedValues.append(genderScores)

# SUSTAINABILITY
susData = dataList("../PACK-HiMCM/PACK HIMCM DATA - Sustainability.csv")
susNegated = [-float(n[1]) for n in susData[1:]]
susScores = [(m-min(susNegated))/(max(susNegated)-min(susNegated)) for m in
                                        susNegated]
normalizedValues.append(susScores)
```

```
# INCLUSIVITY
incData = dataList("../PACK-HiMCM/PACK HIMCM DATA - Inclusivity.csv")
filterData = [float(n[1]) for n in incData[1:] if float(n[1])>=75]
incMin = min(filterData)
incMax = max(filterData)
incScores = [(float(n[1]) - incMin)/(incMax - incMin) for n in incData[1:]]
normalizedValues.append(incScores)

# RELEVANCE & INNOVATION
relInnoData = dataList("../PACK-HiMCM/PACK HIMCM DATA - Relevance & Innovation.
                                        csv")

def convertFloat(lst):
    newLst = []
    for term in lst:
        newLst.append(float(term))
    return newLst

def findPeaks(lst):
    peaks = []
    for i in range(12):
        peaks.append(max(lst[i*12:(i*12+11)]))
    return peaks

def linearRegSlope(lst):
    xAxis = range(0, len(lst))
    m,b = np.polyfit(xAxis, lst, 1)
    return m

slopes = [linearRegSlope(findPeaks(convertFloat(n[1:]))) for n in relInnoData[1:]
                                        ]
relInnoScores = [(m-min(slopes))/(max(slopes)-min(slopes)) for m in slopes]
normalizedValues.append(relInnoScores)

# FAIR-PLAY
dopeData = dataList("../PACK-HiMCM/PACK HIMCM DATA - Fair Play.csv")
dopeRatios = [-float(dopeData[1:][i][1])/float(accData[1:][i][1]) for i in range(
                                        len(dopeData[1:]))]
dopeScores = [(m-min(dopeRatios))/(max(dopeRatios)-min(dopeRatios)) for m in
                                        dopeRatios]
normalizedValues.append(dopeScores)

# SAFETY
safetyData = dataList("../PACK-HiMCM/PACK HIMCM DATA - Safety.csv")
safetyNegated = [-float(n[1]) for n in safetyData[1:]]
safetyScores = [(m-min(safetyNegated))/(max(safetyNegated)-min(safetyNegated))
                                        for m in safetyNegated]
normalizedValues.append(safetyScores)

# For any sports with less than 75 countries/territories in their international
                                        federation, change all scores to -1.

for i in range(len(normalizedValues[0])):
    if normalizedValues[4][i] < 0:
```

```
        print(normalizedValues[4][i])
        for factor in normalizedValues[1:]:
            factor[i] = -1

print(normalizedValues)

results_df = pd.DataFrame(normalizedValues)
results_df.to_csv('../PACK-HiMCM/FinalData.csv', index = False)
```